

GIMLI HARBOUR, MB, CanadaWMO: **715770**Lat: **50.631N**Lon: **96.982W**Elev: **217**StdP: **98.74**Time Zone: **-6.00 (NAC)**Period: **99-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.2	-27.7	-33.7	0.2	-29.9	-31.4	0.2	-27.4	11.5	-10.8	10.5	-11.1	2.7	310	0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	28.0	20.5	26.4	19.8	25.0	19.4	22.9	25.6	21.7	24.6	20.6	23.5	4.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	17.2	24.4	20.7	15.7	23.4	19.6	14.7	22.5	68.9	25.5	64.3	24.7	60.6	23.5	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.8	8.8	-32.6	31.8	2.7	1.7	-34.5	33.0	-36.0	34.0	-37.5	34.9	-39.5	36.1		
(5)			-32.7	25.3	2.6	1.3	-34.6	26.2	-36.1	27.0	-37.6	27.7	-39.5	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-15.5	-14.1	-6.2	1.9	9.6	16.6	20.1	18.9	13.8	6.1	-2.7	-12.2
	DBStd	13.69	7.73	6.97	7.13	5.27	4.81	3.60	2.99	3.20	4.10	4.28	6.06	7.19
	HDD10.0	3501	792	674	503	246	66	1	0	0	12	137	381	688
	HDD18.3	5708	1050	908	762	493	271	73	17	32	145	380	631	947
	CDD10.0	987	0	0	0	2	54	201	313	276	126	16	0	0
	CDD18.3	154	0	0	0	0	2	22	72	50	8	0	0	0
	CDH23.3	737	0	0	0	0	21	99	357	229	31	0	0	0
	CDH26.7	124	0	0	0	0	4	16	59	41	5	0	0	0
Wind	WSAvg	4.2	3.9	3.7	3.9	4.2	4.7	4.2	3.9	4.2	4.6	4.8	4.7	4.0
Precipitation	PrecAvg	545	22	15	24	28	64	92	79	74	59	42	28	23
	PrecMax	751	59	29	75	62	120	195	170	148	117	91	85	55
	PrecMin	358	7	1	3	10	19	38	18	16	19	9	5	6
	PrecStd	96	12	8	15	12	28	42	31	31	26	23	16	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	4.6	11.0	18.5	26.2	28.7	30.8	30.2	26.7	19.7	12.2	3.1
		MCWB	2.1	2.7	7.4	10.2	14.0	19.7	21.9	21.4	17.8	13.3	8.0	1.3
	2%	DB	0.6	2.1	6.9	13.5	21.4	26.1	28.2	27.5	23.5	16.4	8.7	0.7
		MCWB	0.0	0.9	4.4	7.9	14.0	18.0	21.0	20.5	17.8	12.2	6.1	-0.2
	5%	DB	-1.9	-0.5	4.7	10.5	18.4	23.9	26.6	25.7	21.6	14.5	6.6	-0.6
		MCWB	-2.3	-1.4	3.0	7.0	13.1	17.9	20.5	20.1	17.9	11.0	5.0	-1.1
	10%	DB	-4.2	-3.6	2.9	8.2	16.1	22.1	25.3	24.2	19.9	12.5	4.9	-2.1
		MCWB	-4.7	-4.4	1.7	5.7	12.4	17.8	20.1	19.5	16.7	9.6	3.4	-2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	3.1	7.6	10.9	17.5	23.1	25.2	24.0	21.4	14.7	8.6	1.5
		MCDB	3.6	4.3	9.6	15.7	19.5	26.4	27.7	26.3	22.7	17.5	11.0	2.8
	2%	WB	-0.2	1.0	4.8	8.9	15.5	20.7	23.4	22.5	20.1	13.1	6.5	0.0
		MCDB	0.6	2.0	6.6	12.5	18.7	23.4	26.0	25.3	21.5	15.5	8.2	0.7
	5%	WB	-2.5	-1.6	3.0	7.4	14.1	19.5	22.1	21.4	18.6	11.5	5.0	-1.1
		MCDB	-2.0	-0.6	4.5	10.4	16.9	21.9	25.1	24.6	20.7	14.0	6.5	-0.6
	10%	WB	-4.8	-4.5	1.7	5.6	12.7	18.4	21.0	20.4	17.0	9.7	3.5	-2.7
		MCDB	-4.3	-3.8	2.9	8.2	15.9	21.2	24.0	23.4	19.4	12.2	4.8	-2.2
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.4	9.4	8.2	7.9	7.4	8.2	8.3	7.6	6.3	6.1	7.2
		MCDBR	9.4	9.9	9.2	11.4	12.1	10.3	10.0	10.4	10.3	9.4	7.8	7.9
		MCWBR	8.9	8.7	7.0	6.9	6.2	5.5	5.5	5.8	6.1	5.8	5.7	7.4
	5% WB	MCDBR	9.3	9.8	8.8	11.4	9.6	8.3	8.5	8.8	8.1	8.2	7.3	8.0
		MCWBR	8.6	8.7	6.8	7.2	5.6	5.4	5.8	5.8	5.7	5.5	5.5	7.4
Clear-Sky Solar Irradiance	taub	0.264	0.296	0.305	0.354	0.368	0.384	0.394	0.377	0.346	0.301	0.271	0.247	
	taud	2.219	2.137	2.283	2.335	2.373	2.355	2.358	2.406	2.479	2.559	2.420	2.285	
	Ebn at Noon	763	823	898	888	888	872	857	855	848	838	769	737	
	Edh at Noon	70	101	106	114	115	118	116	105	88	67	58	56	
All-Sky Solar Radiation	RadAvg	1.22	2.39	3.67	5.04	5.45	5.79	6.03	5.01	3.55	2.06	1.10	0.89	
	RadStd	0.12	0.17	0.27	0.34	0.46	0.46	0.48	0.35	0.32	0.27	0.16	0.09	